Assignment
Practical and oral examination

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Group
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Course code
GDT2JM

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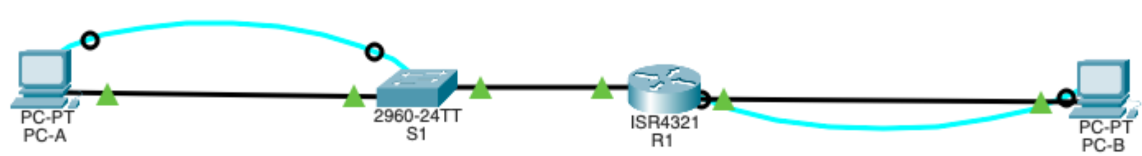
## 1. Introduction

The purpose of this assignment was to build and configure a small routed network using Cisco Packet Tracer. The task included configuring a router, a switch and two PCs to achieve end-to-end connectivity between two separate IP networks. The network consists of one router (R1), one switch (S1) and two hosts (PC-A and PC-B). The router connects two different subnets and enables communication between them.

## 2. Topology




The topology consists of:

- R1 (Cisco 4321 router)
- S1 (Cisco 2960 switch)
- PC-A connected to S1
- PC-B connected to R1
- Proper Ethernet cabling between devices

All devices are clearly labeled.

3. Addressing Overview

Router (R1)

- G0/0/0 → 192.168.0.1 /24 (Connected to Switch / PC-A LAN)
- G0/0/1 → 192.168.1.1 /24 (Connected to PC-B)

Switch (S1)

- VLAN 1 → 192.168.0.2 /24
- Default Gateway→ 192.168.0.1

PC-A

- IP Address: 192.168.0.3
- Subnet mask: 255.255.255.0
- Default Gateway: 192.168.0.1

PC-B

- IP Address: 192.168.1.3
- Subnet mask: 255.255.255.0
- Default Gateway: 192.168.1.1

4. Configuration Evidence

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   192.168.0.1     YES manual up                    up
GigabitEthernet0/0/1   192.168.1.1     YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```

Router – R1

```
R1#show running-config | section GigabitEthernet
interface GigabitEthernet0/0/0
 description To Switch / PC-A
 ip address 192.168.0.1 255.255.255.0
 duplex auto
 speed auto
interface GigabitEthernet0/0/1
 description To Pc-B
 ip address 192.168.1.1 255.255.255.0
 duplex auto
 speed auto
R1#show running-config | include hostname
hostname R1
R1#show running-config | include enable
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
R1#show running-config | include banner
banner motd ^CUnauthorized access to this device is prohibited!^C
```

Switch – S1

```
S1#show running-config | section Vlan1
interface Vlan1
 ip address 192.168.0.2 255.255.255.0
S1#show running-config | include hostname
hostname S1
S1#show running-config | include default-gateway
ip default-gateway 192.168.0.1
```

## 5. Connectivity Verification

The following connectivity tests were successfully verified:

- PC-A → PC-B (192.168.1.3)

```
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=127
Reply from 192.168.1.3: bytes=32 time<1ms TTL=127
Reply from 192.168.1.3: bytes=32 time<1ms TTL=127
Reply from 192.168.1.3: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

- PC-A → Router (192.168.0.1)

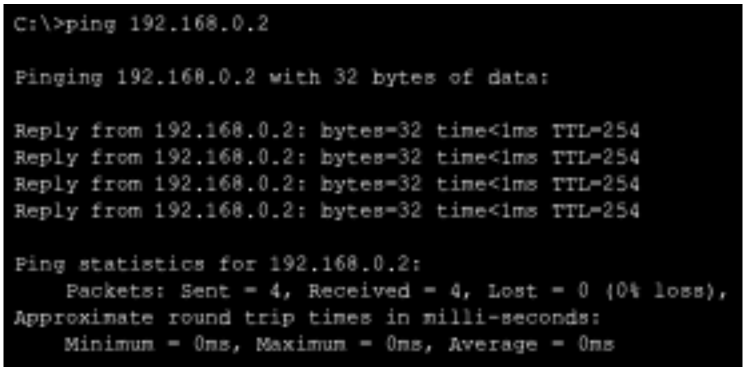
```
C:\>ping 192.168.0.1

Pinging 192.168.0.1 with 32 bytes of data:

Reply from 192.168.0.1: bytes=32 time=1ms TTL=255
Reply from 192.168.0.1: bytes=32 time<1ms TTL=255
Reply from 192.168.0.1: bytes=32 time<1ms TTL=255
Reply from 192.168.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

- PC-B → Switch (192.168.0.2)



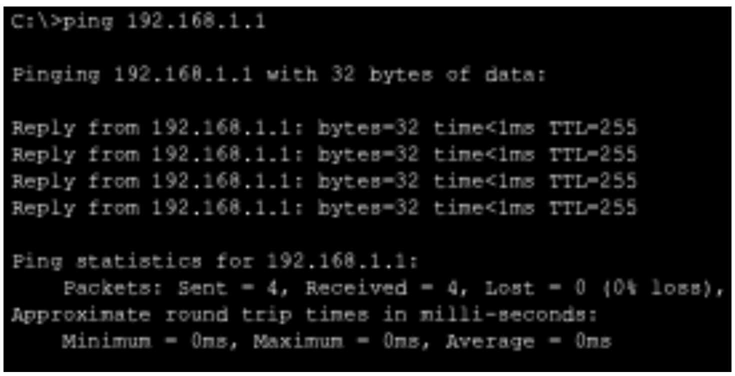
```
C:\>ping 192.168.0.2

Pinging 192.168.0.2 with 32 bytes of data:

Reply from 192.168.0.2: bytes=32 time<1ms TTL=254
Reply from 192.168.0.2: bytes=32 time<1ms TTL=254
Reply from 192.168.0.2: bytes=32 time<1ms TTL=254
Reply from 192.168.0.2: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

- PC-B → Router (192.168.1.1)



```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

All pings returned successful replies, confirming proper IP configuration, routing functionality and VLAN configuration.

## 6. Reflection

This configuration of both the router and switch worked as excepted once the correct interface assignments were identified. The most challenging part was identifying that the router interfaces were connected to different physical ports than initially assumed, which required troubleshooting using the `show cdp neighbors` command. Through this lab, I learned the importance of verifying physical connections and understanding how interface configuration must match the actual topology. The lab strengthened my understanding of routing between subnets and VLAN-based switch management.